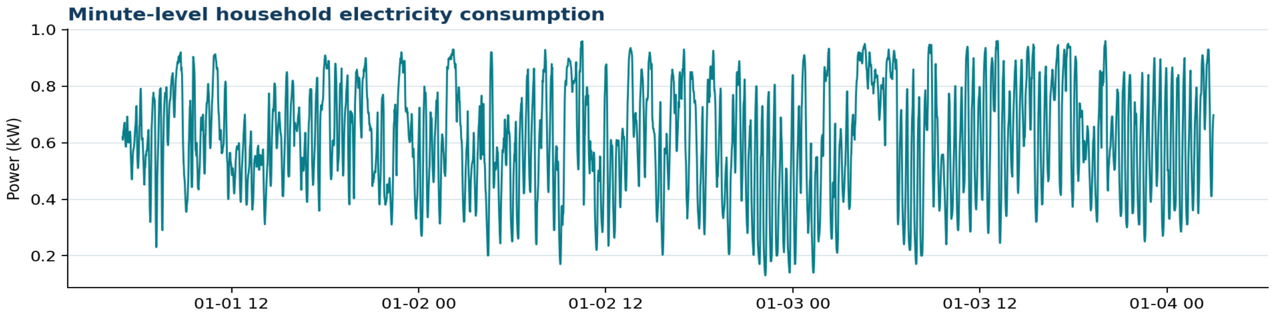
Use Case 2 - Smart-Home Energy Forecasting

Three-minute-ahead household electricity forecasting at one-minute resolution

Dataset identity and quality

Checked CSV	use_case_2_smart_home_energy.csv
Source	Kaggle - Smart Home Dataset with Weather Information
Rows / target	4,200 timestamps / use_kW
Covered period	2016-01-01 05:00 to 2016-01-04 02:59
Cadence	One minute, complete sequence
Range / mean	0.13 to 0.96 kW; mean 0.611 kW
Missing / duplicates	0 missing timestamps; 0 missing values; 0 duplicates



Use-case role

The source collection contains whole-home and appliance-level consumption together with weather variables. This ML2++ validation subset uses only the total household-consumption variable originally named use [kW], renamed use_kW for the DSL model.

Train/test and forecast design

Use a chronological 80/20 split: 3,360 observations for fitting and 840 for the held-out period. With complete three-step targets, the evaluation has 838 valid test origins. Do not shuffle. The outputs are t+1, t+2, and t+3 minutes.

Models

ARIMA(1,1,1) is evaluated as a fixed non-seasonal model. Additive Holt-Winters uses a damped additive trend and an additive seasonal period of 60 observations, representing an hourly cycle at minute resolution. AutoML and hyperparameter search are off.

Interpretation boundary

The subset spans about 70 hours, which is too short for a defensible 1,440-minute daily seasonal evaluation. The period-60 Holt-Winters configuration is a technical validation, not evidence that an hourly seasonal structure is optimal for household demand.